

Individual Differences & Task Attention in Cybersickness: A Call for a Standardized Approach to Data Sharing

Stephen B. Gilbert*, Angelica Jasper, Nathan C. Sepich,
Taylor A. Doty, Jonathan W. Kelly, Michael C. Dorneich

Iowa State University

ABSTRACT

Cybersickness research has lacked a focus on individual differences, particularly in the phases of cybersickness after initial sensitivity: the rate of adaptation during the stimulus, the rate of recovery after the stimulus, and the rate of habituation. Additionally, research has neglected the exploration of the impact of a task's attentional load on cybersickness. Finally, addressing these gaps within the cybersickness research community could be more effective if more standardized, "menu-driven" experimental protocols could be agreed upon, as well as a set of reporting standards to expedite a deeper understanding of cybersickness.

Keywords: Cybersickness, individual differences, cognitive load, attention.

Index Terms: [Human-centered computing: Human computer interaction (HCI): Interaction paradigms]: Virtual reality; [Human-centered computing: Human computer interaction (HCI)]: Empirical studies in HCI

1 INTRODUCTION

Visually Induced Motion Sickness (VIMS), or cybersickness, is a type of motion sickness stemming from the perception of motion while not moving [1], which can occur frequently when using virtual reality [2]. This paper suggests approaches to two future research areas—the role of individual differences and attentional demands on cybersickness, as well as a method of standardization that may aid researchers in the community.

Multiple researchers have noted that cybersickness is influenced by a combination of characteristics of the person and of the stimulus [3]. A smaller number of studies have noted that the task also has an effect [3, 4]. More recently, researchers have begun to build predictive models that could potentially address the interactions between these characteristics [e.g., 5]. Figure 1 attempts to illustrate those potential interactions. The effects of the three main components are described briefly below: individual differences, stimulus perception, and task attention.

1.1 The Individual

Per Lackner [6], individuals vary significantly in their susceptibility to cybersickness at three stages: the initial sensitivity to becoming sick, the rate of real-time adaptation during the experience of the stimulus, and the rate of recovery after the stimulus. The second stage, adaptation, may be counterintuitive, but some people can adapt to a stimulus and after initial feeling of sickness become less sick over time rather than more [6]. A fourth "stage" could be considered the habituation over time after repeated

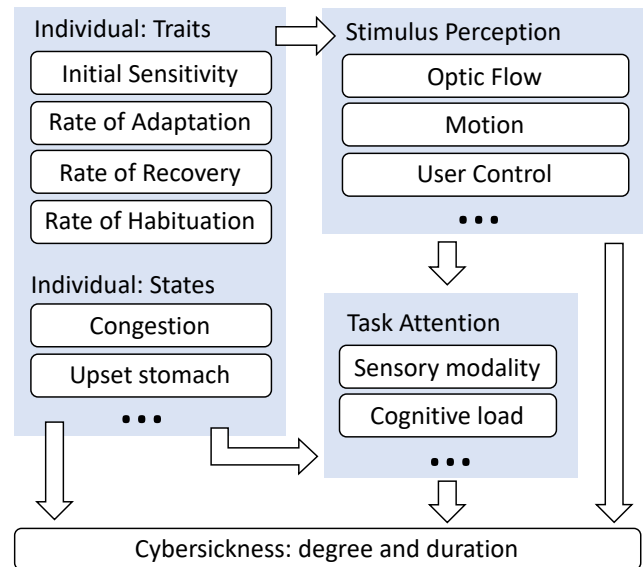


Figure 1: A hypothesized model of factors affecting each other and cybersickness.

exposure. Results are mixed on whether habituation transfers from stimulus to stimulus and task to task [7, 8], and overall there is less research exploring individual differences in habituation.

Different combinations of the levels of these stages can lead to very different experiences of cybersickness over time. For instance, Person A, with low sensitivity, low adaptation, long recovery time, and high rate of habituation might suffer less severely from cybersickness than Person B with high sensitivity, high adaptation, short recovery time, and low rate of habituation. These factors could be said to be traits of the individual, which are present for a longer period of time in the individual's life. Individual differences in initial sensitivity (Stage 1) have been found to be correlated with other individual factors such as age, gender, video game experience [9, 10], and sensitivity to visual cues [11], though less research has focused on individual differences within the other two stages: traits that affect recovery, or the ability to take advantage of different cybersickness mitigation techniques. It is also worth noting that in addition to the longer term traits, there are shorter term states that affect susceptibility, such as illness, congestion, and upset stomach [9]. All of these factors relate to characteristics of the individual, and if better understood, VR experienced could be personalized and adapted to individuals' needs. **Increasing research in individual differences, in trait stages 2, 3, 4, and individuals' states, is one of the key topic areas of proposed research.**

* gilbert@iastate.edu

1.2 Perception of the Stimulus

Characteristics of the stimulus also affect cybersickness [e.g., 3, 11, 12], such as how much optic flow is present, but it is worth noting that a physical description of the stimulus in terms of contrast, luminance, spatial frequency, etc., is insufficient to predict its impact on sickness. The perception of that stimulus (which may vary between individuals) is also a key factor. Vection, for example, the illusory perception of self-motion, is affected by top-down cognitive factors [13]. Also, one's sense of presence in an environment (a percept) has been shown to be inversely correlated with cybersickness [14, 15], a relationship meriting further exploration, since it is not clear whether sickness causes lower presence or lower presence increases sickness.

The method of stimulus presentation can also affect perception of the stimulus as well. Several of the cybersickness mitigation methods presented in GingerVR [16], for example, include peripheral blurring, dynamic field of view, and other filters for the original stimulus. Also related to presentation, the simple evolution of display hardware may affect presentations dramatically enough that stimuli presented ten years ago, for example, may be difficult to compare with stimuli today.

1.3 Attentional Task Demands

A relatively unexplored topic is the effect of the task performed by the participant, which draws on both the individual's capabilities and the stimulus. While a small number of researchers have noted that the task has an effect [3, 4], they focus on task factors like duration, amount of user control, and sitting vs. standing. However, given phenomena such as inattentional blindness, in which participants do not notice significant changes in their environment while their attention is focused on a task [17, 18], and research on distracted driving [e.g., 19], it is likely that the attention and cognitive load required by the task given to participants may affect cybersickness. Based on this research, the authors hypothesize that some tasks that require significant cognitive or visual attention may "distract" from cybersickness. *The effect on cybersickness of the task and the attention it requires is the second key topic area of proposed research.*

2 APPROACH: A "MENU-BASED" RESEARCH PROTOCOL

To address these, we propose the overall research approach shown in Figure 2. It is not dissimilar from many typical experimental designs: participants are recruited, they complete a task in a virtual environment designed to induce cybersickness, they experience mitigation methods either real-time during the task or afterwards, and (if desired) the experience is repeated after passing of time to measure habituation. Measurement of cybersickness occurs at multiple points: an initial baseline, regularly during the task, after the task, and after mitigation is complete. In the more generalized version, a researcher could choose from a public menu of options for each component or add a new option to the menus. This structured approach allows for systematic varying of the variables discussed above: the states and traits of participants, the stimuli, the task, the mitigation techniques, and the habituation schedule, as well as the method of measuring cybersickness.

The authors suggest that if this approach were more standardized via menu options, comparisons between tasks, stimuli, mitigations, demographic groups, and cybersickness measures could be more systematic. E.g., Research Group A, that wants to explore the impact different mitigation techniques, would decide to keep their other variables constant based on Research Group B's previous work, i.e., the participant pool, the task, the timings, the cybersickness measures, etc. This approach would make a statistical comparison between Group A's studies and Group B's studies more feasible, allowing Group A to not only compare the

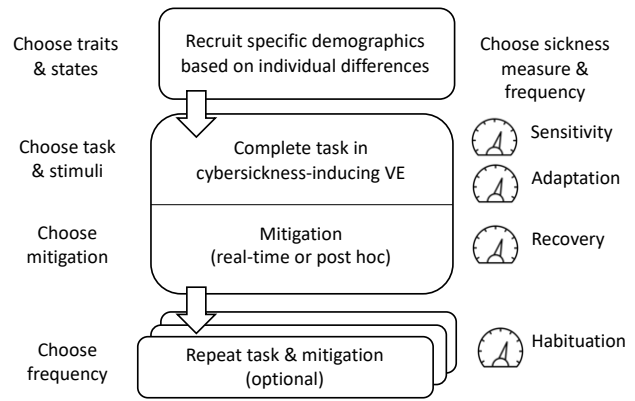


Figure 2. Proposed "menu-based" research approach: adopt a standardized experimental structure with the option to choose options from public menus of varied traits and states of participants, varied tasks and stimuli, varied mitigation methods, and (if desired) varied frequency of habituation trials. Researchers could also add options to the menus when desired. This approach would allow a large database of de-identified data to be shared across researchers more easily.

effect of their own mitigation methods with each other but also compare their interventions with Group B's mitigation methods.

Also, resulting data from cybersickness researchers worldwide could in some cases be pooled, resulting in stronger statistical power and faster discovery of the underlying characteristics that affect cybersickness. A related vision for broader collaboration within the VR community and the related challenges thereof are expressed well by Steed et al. [20].

2.1 Crowdsourcing and Data Sharing

Crowdsourcing data, in which typically a single research team solicits volunteered data from a large sample of the general public, is by no means new. Computer vision researchers successfully crowdsourced home videos of day-to-day tasks to aid in machine learning [21]. Players of the game Foldit generated novel results in protein folding [22]. Players of the game Sea Hero Quest generated data about the spatial navigation abilities of millions of participants [23]. Crowdsourcing can take place with participant recruitment as well as data collection; members of the Immersive Experiences Working Group at Queen Mary, University of London recently created the XRDRN website (xrdrn.org), which allows multiple researcher groups to share in the participant recruiting process by posting XR studies there.

What goes beyond crowdsourcing is true data sharing – multiple research teams agreeing on standardized data collection approach and pooling those data. It is worth noting that aligning the incentives and standards for sharing data successfully can be quite difficult. The biomedical research community has been attempting this for several decades with mixed results. For example, the U.S. National Cancer Institute launched caBIG in 2004 as a future hub of all cancer research data, an effort that included software tools to aid in data collection and analysis. In 2011, while some progress was noted, caBIG was deemed a failure overall and ended [24]. The reasons were complex, related to software standards, industry competition, and incentives to participate [25]. That said, efforts to standardize medical data formatting (e.g., FHIR, Fast Healthcare Interoperability Resources) and data sharing requirements (e.g., USCDI, the United States Core Data for Interoperability) continue so that medical researchers and medical practitioners can benefit more quickly from each other's work.

An example of more successful data sharing can be seen in the work of Audrey Rorrer and collaborators, who created a toolkit that standardized a method of program evaluation of National Science Foundation Research Experience for Undergraduates summer experiences within the NSF CISE directorate (reu.uncc.edu/cise-reu-toolkit) [26]. By including toolkit features that eased the burden of principal investigators, adoption of the toolkit grew and enabled Rorrer's team to draw insights from a national dataset [26]. When data collection for this toolkit began in 2010, it was not yet as customary to share de-identified data online, so the IRB protocols were not in place to do so with data collected. However, if data were collected starting today, one can envision the national dataset being available on Open Science Framework to facilitate analyses by multiple labs. As noted by Steed et al. [20], regulations such as GDPR can make this process more complicated, but not impossible.

If a toolkit could be created for a menu-based cybersickness research protocol that made the research process simpler for researchers, researchers might be more willing to adopt it. These benefits might include:

- IRB templates
- Survey templates (e.g., "Please use at least these questions but add your own if desired")
- Standardized definitions for terms like "vection" and "cybersickness"
- Open-source repository of VEs for tasks
- Open-source repository of code for mitigation techniques
- Standardized tools for measuring cybersickness
- Open-source repository of R data analysis scripts
- Research reporting guidelines that ensure that all publications report a minimal set of required information.

This last benefit is based on the approach used with biomedical clinical trials, in which publications of results are standardized according to reporting guidelines such as CONSORT [27] (see www.equator-network.org for others). Some journals require that all submitted articles follow the CONSORT requirements, for example, which include making sure that authors describe the participant recruitment method, the trial design, the sample size, the statistics used, etc. A good article will offer this information by default, but having reporting guidelines can ensure that researchers who later read our publications will have the information they need to perform the broader statistical comparisons described above.

3 EXAMPLE: A CORN MAZE & FOUR MITIGATION TECHNIQUES

To illustrate an example of this standardized approach, we describe how a previous study on mitigation techniques [10] could be described in these terms.

3.1 Participant Recruitment & Allocation

Participants in this example study were 57 undergraduates at the authors' university. Because the authors noted 21 females and 36 males, we can infer that recruiting did not use blocking to balance participants by gender. However, given the menu-based approach in Figure 2, one could imagine that researchers conducting a similar future study might choose to recruit based on variables related to individual differences, e.g., age, sex, gender, video game experience, visual cue sensitivity, etc.

This study's experimental design included two independent variables: movement control (two conditions) and mitigation technique (four conditions). For participant allocation to conditions, the authors noted that participants were randomly assigned. Having reporting standards as mentioned above would likely guide the authors to report more specifically: was it simple randomization, block randomization, or stratified randomization? Reporting guidelines would also encourage researchers to publish the exact demographics they recruited to promote appropriate data

pooling. Stating these details would ease the process of another researcher conducting a systematic review to include this paper in a meta-analysis.

3.2 Stimulus and Task

The example study used the Cybersickness Corn Maze VE (github.com/isuvrac/CyberSickness-Cornmaze) as the stimulus. The Corn Maze was developed as a testbed for inducing cybersickness based on tasks from the Virtual Environment Performance Assessment Battery (VEPAB) [28]. Tasks within the maze include 44 left and right 90 degree turns, trampolines, optokinetic drums, ramps, and spiral slides [29]. Settings within the maze can be adjusted to induce more severe symptoms (e.g., no movement control or higher speed), allowing for fine-tuned control and experimental variation to test different combinations of conditions. In the proposed more generalized research approach, one can imagine a webpage listing an array of open-sourced VEs and their specifications, so that a researcher who wanted to focus on rotational vection, for example, could choose an appropriate VE. Ideally, all of the VEs vetted for that webpage would have the ability to output similar data logs that align to a cybersickness data standard.

The task in the example study was simply to endure the Corn Maze, effectively the "null task." However, in this VE and others, one can envision tasks chosen specifically to probe their impact on cybersickness, e.g., a visual task such as spotting certain objects in the maze, a more cognitive task such as the n-back task, or perhaps a psychomotor task of aiming a virtual laser at objects along the way. In the standard approach, a list of tasks (and the code to implement them) could be cataloged, along with a rating of the mental workload required by each.

3.3 Mitigation and Cybersickness Measurement

In the example study, after participants grew sick in the Corn Maze, they experienced one of four mitigation activities. Generalized reporting guidelines could be created that would ensure that whenever mitigation techniques are used in other studies, all appropriate timings are reported in publications. Mitigation techniques can also occur simultaneously with the stimulus, e.g., the AuthenticNose or DynamicGaussianBlur effects included in GingerVR [16]. Similar to the idea of a webpage of VEs for cybersickness research, mitigation techniques could be listed, along with their parameters so that one paper that that uses DynamicGaussianBlur might consider using similar parameter settings to other studies unless those parameters were desired as independent variables.

The example study used the Simulator Sickness Questionnaire (SSQ) [30] to measure cybersickness before the maze, during the maze, during the mitigation, and at the end. However, researchers have administered the SSQ in a variety of ways, e.g., verbally, on paper, via an online survey, yielding results that are difficult to compare [31]. A more generalized approach would provide templates and standard approaches for this so that studies using SSQ can reasonably compare data with each other. Of course there are other measures of cybersickness as well that could be included in a menu of methods.

3.4 Data Analysis

The example study uses hypothesis testing techniques to compare participants' results. Since statistical power analyses for hypothesis testing, used to determine how many participants are required for a study, are based on a desire to detect a certain size difference between the means of two groups, the question arises, "How much difference in sickness should I expect to see in my study?" Adopting a more generalized approach within our community might enable the publication of difference thresholds for each of

the cybersickness measures, essentially creating templates for power analysis.

Regression based analysis approaches such as multilevel models and structural equation modeling (SEM) might be preferred if seeking to clarify the relationships between the various factors in Figure 1. Using this approach would allow one to examine the potential mediating effects of task attention on the relationship between stimulus and sickness, as well as individual characteristics and sickness, while also accounting for direct relationships. While these approaches sometimes require more data than hypothesis testing, in a more generalized data-sharing approach, the model can be iteratively refined as more data arrives.

4 SUMMARY

Based on gaps in previous research, this paper calls for more research focused on individual differences that affect all phases of cybersickness, as well as additional research on the impact of a task's attentional load on cybersickness. In addition, establishing a strong community of researchers who agree to a more standardized experimental protocol and set of reporting standards would hasten the discovery of methods to mitigate and prevent cybersickness.

REFERENCES

- [1] R. S. Kennedy, J. Drexler, and R. C. Kennedy, "Research in visually induced motion sickness," *Applied ergonomics*, vol. 41, no. 4, pp. 494-503, 2010, doi: 10.1016/j.apergo.2009.11.006.
- [2] T. Lewis. (2015, March 13) When will virtual-reality headsets stop making people sick? *Live Science*. Available: <https://www.livescience.com/50129-virtual-reality-nausea-sickness.html>
- [3] S. Davis, K. Nesbitt, and E. Nalivaiko, "A Systematic Review of Cybersickness," presented at the Proceedings of the 2014 Conference on Interactive Entertainment, Newcastle, NSW, Australia, 2014.
- [4] E. M. Kolasinski, *Simulator sickness in virtual environments*. US Army Research Institute for the Behavioral and Social Sciences, 1995.
- [5] L. Rebenitsch and C. Owen, "Estimating cybersickness from virtual reality applications," *Virtual Reality*, pp. 1-10, 2020.
- [6] J. R. Lackner, "Motion sickness: more than nausea and vomiting," *Experimental Brain Research*, vol. 232, no. 8, pp. 2493-2510, 2014, doi: 10.1007/s00221-014-4008-8.
- [7] J. F. Golding, "Motion sickness susceptibility," *Autonomic Neuroscience*, vol. 129, no. 1, pp. 67-76, 2006, doi: 10.1016/j.autneu.2006.07.019.
- [8] A. Graybiel, A. B. Thompson, F. R. Deane, A. R. Fregly, J. K. Colehour, and E. L. Ricks, *Transfer of habituation of motion sickness on change in body position between vertical and horizontal in a rotating environment*. Naval Aerospace Medical Institute, Naval Aerospace Medical Center, 1968.
- [9] J. J. LaViola Jr., "A discussion of cybersickness in virtual environments," *ACM Sigchi Bulletin*, vol. 32, no. 1, pp. 47-56, 2000.
- [10] A. Jasper, N. Cone, C. Meusel, M. Curtis, M. C. Dorneich, and S. B. Gilbert, "Visually Induced Motion Sickness Susceptibility and Recovery Based on Four Mitigation Techniques," *Frontiers in Virtual Reality*, vol. 1, p. 22, 2020, doi: 10.3389/frvir.2020.582108.
- [11] K. Stanney *et al.*, "Identifying Causes of and Solutions for Cybersickness in Immersive Technology: Reformulation of a Research and Development Agenda," *International Journal of Human-Computer Interaction*, vol. 36, no. 19, pp. 1783-1803, 2020/11/25 2020, doi: 10.1080/10447318.2020.1828535.
- [12] L. Rebenitsch and C. Owen, "Review on cybersickness in applications and visual displays," *Virtual Reality*, vol. 20, no. 2, pp. 101-125, 2016, doi: 10.1007/s10055-016-0285-9.
- [13] S. Palmisano, R. S. Allison, M. M. Schira, and R. J. Barry, "Future challenges for vection research: definitions, functional significance, measures, and neural bases," (in English), *Frontiers in Psychology*, Review vol. 6, no. 193, 2015-February-27 2015, doi: 10.3389/fpsyg.2015.00193.
- [14] A. Maneuvrier, L. M. Decker, H. Ceyte, P. Fleury, and P. Renaud, "Presence Promotes Performance on a Virtual Spatial Cognition Task: Impact of Human Factors on Virtual Reality Assessment," (in English), *Frontiers in Virtual Reality*, Original Research vol. 1, no. 16, 2020-October-29 2020, doi: 10.3389/frvir.2020.571713.
- [15] S. Weech, S. Kenny, and M. Barnett-Cowan, "Presence and cybersickness in virtual reality are negatively related: a review," *Frontiers in psychology*, vol. 10, p. 158, 2019.
- [16] S. Ang and J. Quarles, "GingerVR: An Open Source Repository of Cybersickness Reduction Techniques for Unity," in *2020 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*, 2020: IEEE, pp. 460-463.
- [17] D. J. Simons and C. F. Chabris, "Gorillas in our midst: Sustained inattention blindness for dynamic events," *perception*, vol. 28, no. 9, pp. 1059-1074, 1999.
- [18] D. J. Simons and D. T. Levin, "Failure to detect changes to people during a real-world interaction," *Psychonomic Bulletin & Review*, vol. 5, no. 4, pp. 644-649, 1998.
- [19] J. D. Lee, "Dynamics of driver distraction: The process of engaging and disengaging," *Annals of advances in automotive medicine*, vol. 58, p. 24, 2014.
- [20] A. Steed *et al.*, "Evaluating immersive experiences during Covid-19 and beyond," *interactions*, vol. 27, no. 4, pp. 62-67, 2020, doi: 10.1145/3406098.
- [21] G. A. Sigurdsson, G. Varol, X. Wang, A. Farhadi, I. Laptev, and A. Gupta, "Hollywood in homes: Crowdsourcing data collection for activity understanding," in *European Conference on Computer Vision*, 2016: Springer, pp. 510-526.
- [22] S. Cooper *et al.*, "Predicting protein structures with a multiplayer online game," (in eng), *Nature*, vol. 466, no. 7307, pp. 756-760, 2010, doi: 10.1038/nature09304.
- [23] A. Coutrot *et al.*, "Global Determinants of Navigation Ability," *Current Biology*, vol. 28, no. 17, pp. 2861-2866.e4, 2018.
- [24] Board of Scientific Advisors Ad Hoc Working Group, "An Assessment of the Impact of the NCI Cancer Biomedical Informatics Grid (caBIG)," National Institutes of Health: National Cancer Institute, 2011.
- [25] S. Leonelli, "Global data for local science: Assessing the scale of data infrastructures in biological and biomedical research," *BioSocieties*, vol. 8, no. 4, pp. 449-465, 2013/12/01 2013, doi: 10.1057/biosoc.2013.23.
- [26] A. S. Rorrer, J. Allen, and H. Zuo, "A national study of undergraduate research experiences in computing: Implications for culturally relevant pedagogy," in *Proceedings of the 49th ACM Technical Symposium on Computer Science Education*, 2018, pp. 604-609.
- [27] K. F. Schulz, D. G. Altman, and D. Moher, "CONSORT 2010 statement: updated guidelines for reporting parallel group randomised trials," *Trials*, vol. 11, no. 1, pp. 1-8, 2010.
- [28] D. R. Lampton, B. W. Knerr, S. L. Goldberg, J. P. Bliss, J. M. Moshell, and B. S. Blau, "The Virtual Environment Performance Assessment Battery (VEPAB): Development and Evaluation," *Presence: Teleoperators and Virtual Environments*, vol. 3, no. 2, pp. 145-157, 1994, doi: 10.1162/pres.1994.3.2.145.
- [29] M. K. Curtis *et al.*, "Mitigating Visually Induced Motion Sickness A virtual hand-eye coordination task," *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*, vol. 59, no. 1, pp. 1839-1843, 2015.
- [30] R. S. Kennedy, N. E. Lane, K. S. Berbaum, and M. G. Lilienthal, "Simulator Sickness Questionnaire: An Enhanced Method for Quantifying Simulator Sickness," *The International Journal of Aviation Psychology*, vol. 3, no. 3, pp. 203-220, 1993, doi: 10.1207/s15327108ijap0303_3.
- [31] P. Bimburg, T. Weissker, and A. Kulik, "On the Usage of the Simulator Sickness Questionnaire for Virtual Reality Research," in *IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops: Workshop on Immersive Sickness Prevention (WISP)*, Virtual, 2020.